

CAB-Igbo Human Evaluation Guidelines

1 Annotator Guidelines

1.1 Background

The CAB-Igbo benchmark evaluates how AI models represent and apply Igbo cultural knowledge. In this study, trained annotators assess model-generated responses to cultural questions using a standardised rubric. The evaluation has two objectives:

- validate the automated LLM-as-Judge system, and
- compare model performance across persona conditions.

Model identities remain hidden throughout the evaluation. Annotators will only see anonymised labels such as *Model A* and *Model B*.

1.2 Evaluation Setup

Annotators will evaluate 50 questions sampled from the CAB-Igbo dataset. Each question has responses from 8 models under 11 experimental conditions, producing 4,400 evaluation rows.

1.2.1 Question Types

- **Type 1:** English question, English answer
- **Type 2:** English question, Igbo answer
- **Type 3:** Igbo question, Igbo answer

1.2.2 Conditions

- Baseline (no persona)
- Imo persona conditions
- Ebonyi persona conditions

The persona categories are:

- Young Teenage Boy
- Married Woman

- Married Man
- Widowed Woman
- Professor

The benchmark covers domains including funerals, religion, food, festivals, and social norms.

1.3 Scoring Constructs

Each response is scored on four constructs using a 1–5 integer scale.

1.3.1 Epistemic Fidelity (EF)

Measures the factual accuracy and cultural reliability of the response.

- 5 = Completely accurate and culturally specific
- 4 = Mostly accurate with minor imprecision
- 3 = Partially accurate with noticeable errors
- 2 = Mostly inaccurate or misleading
- 1 = Completely incorrect or fabricated

1.3.2 Representational Richness (RR)

Measures the depth and nuance of cultural representation.

- 5 = Rich, nuanced, and culturally grounded
- 4 = Good depth with minor gaps
- 3 = Basic or surface-level explanation
- 2 = Minimal or vague cultural detail
- 1 = No meaningful cultural content

1.3.3 Linguistic Quality (LQ)

Measures fluency, grammar, and language use.

- 5 = Fluent and grammatically correct
- 4 = Minor grammatical issues
- 3 = Understandable despite noticeable errors
- 2 = Frequent errors that reduce readability
- 1 = Largely unintelligible

For Igbo responses, annotators should assess Igbo grammar and morphology directly.

1.3.4 Pragmatic Proficiency (PP)

Measures contextual appropriateness and communicative relevance.

- 5 = Fully appropriate and context-aware
- 4 = Mostly appropriate with minor inconsistencies
- 3 = Partially appropriate
- 2 = Largely inappropriate or contextually weak
- 1 = Completely misaligned or off-topic

The total response score is calculated as:

$$\text{Total Score} = EF + RR + LQ + PP$$

The total score ranges from 4 to 20.

1.4 Annotation Procedure

Annotators should complete the following steps:

1. Open the file `CAB_Igbo_Human_Evaluation_Sheet.xlsx`.
2. Go to the `Annotation Sheet` tab.
3. For each row:
 - (a) Read the question carefully.
 - (b) Review the prompt type and condition.
 - (c) Read the model-generated response.
 - (d) Assign integer scores (1–5) for EF, RR, LQ, and PP.
 - (e) Add comments where clarification is necessary.
4. Ensure all rows are completed.
5. Save the file regularly.

1.5 Evaluation Rules

- Evaluate each response independently.
- Responses marked [No response found] or left blank should receive a score of 1 for all constructs.
- Assess Igbo fluency under Linguistic Quality for Type 2 and Type 3 tasks.
- Do not discuss scores with other annotators before completing the evaluation.
- If uncertain between two scores, assign the lower score and provide a brief comment.

1.6 Inter-Rater Reliability

To ensure consistency, two annotators will independently score overlapping subsets of the dataset. Agreement will be measured using weighted Cohen's Kappa. The target reliability threshold is:

$$\kappa \geq 0.60$$

This threshold indicates strong inter-rater agreement.